

MATH3171 (Mathematical Biology III)
YEAR 2025–26, MICHAELMAS TERM

PROBLEM SHEET 4

This problem sheet is due in by **5pm** on **Wednesday 10 December 2025**. You may submit it on paper to my office (MCS 3008) or electronically on Gradescope.

Topics covered: Chapter 6.

This whole problem sheet concerns the version of the spatial predator–prey system for two populations, $u(x, t)$ and $v(x, t)$, that we saw in class:

$$\begin{aligned}\frac{\partial u}{\partial t} &= u(1 - u - v), & (\text{👴}) \\ \frac{\partial v}{\partial t} &= \frac{\partial^2 v}{\partial x^2} + \alpha v(u - \beta), & (\text{🎄})\end{aligned}$$

where the constants $\alpha, \beta > 0$.

In class we stated that the homogeneous system has three equilibria,

- $(0, 0)$, a saddle point,
- $(1, 0)$, a saddle point,
- $(\beta, 1 - \beta)$, a stable node when $4\alpha < \beta/(1 - \beta)$, and a stable spiral for $4\alpha > \beta/(1 - \beta)$,

and that this suggests possible trajectories between our equilibria:

from $(0, 0)$ to $(\beta, 1 - \beta)$ and from $(1, 0)$ to $(\beta, 1 - \beta)$.

1. (a) Draw a phase plane of the homogeneous system: plot v against u , marking on the equilibria, nullclines, and some sample trajectories.
(b) Recall that $(0, 0)$ is a saddle point. Look at trajectories leaving $(0, 0)$. What does this suggest about the existence of a trajectory from $(0, 0)$ to $(\beta, 1 - \beta)$?
2. We now look for travelling wave solutions to the full system from $(1, 0)$ to $(\beta, 1 - \beta)$. We place the stable equilibrium at $x = -\infty$ and the unstable equilibrium at $x = \infty$, and look for rightwards-travelling waves of the form $z = x - ct$, with $c > 0$.
 - (a) If we instead placed the stable equilibrium at $x = \infty$ and the unstable equilibrium at $x = -\infty$, what form of z would we use?
 - (b) Show that, with $z = x - ct$ and $c > 0$, and by introducing $w = dv/dz$, we arrive at the three-dimensional system,

$$\begin{aligned}\frac{du}{dz} &= \frac{u}{c}(u + v - 1), & (\text{🦌}) \\ \frac{dv}{dz} &= w, & (\text{❄️}) \\ \frac{dw}{dz} &= \alpha v(\beta - u) - cw, & (\text{🍁})\end{aligned}$$

and write down the boundary conditions as $z \rightarrow \pm\infty$.

3. We are now going to perform a linear stability analysis on the equilibrium at $(u, v, w) = (1, 0, 0)$.

- (a) Write down the 3×3 Jacobian of the system (🦌 ❄️ 🍁).
- (b) Find the eigenvalues evaluated at $(1, 0, 0)$.
- (c) Make an argument for why we cannot have complex eigenvalues and thus derive a constraint on the squared wavespeed, c^2 .

4. We now look at the stability of the other equilibrium, $(\beta, 1 - \beta, 0)$.

- (a) Evaluate the Jacobian at this equilibrium and show that the resulting eigenvalue equation is

$$\lambda^3 + \lambda^2 \left(c - \frac{\beta}{c} \right) - \beta \lambda + \frac{\beta}{c} \alpha (1 - \beta) = 0. \quad (\text{🕯️})$$

- (b) This is a positive cubic polynomial, so we might expect there to be a local minimum and maximum. Sketch the polynomial and show that the local minimum and maximum are at

$$\lambda_{\max}, \lambda_{\min} = -\frac{1}{3} \left[c - \frac{\beta}{c} \pm \sqrt{\left(c - \frac{\beta}{c} \right)^2 + \text{👴}} \right], \quad (\text{🍪})$$

where you should state the value of 👴. (Just to reiterate: these are *not* necessarily eigenvalues, just the local maximum and minimum of the eigenvalue equation.)

- (c) See that these values of the local maximum and minimum of the eigenvalue equation do not depend on the parameter α . This suggests that increasing α just corresponds to shifting the cubic upwards. Convince yourself that this is the case. Are you convinced?
- (d) Consider eq. (🕯️), then, with $\alpha = 0$ (its lowest possible value), and find the three real eigenvalues which satisfy this equation.
- (e) Therefore we have that when $\alpha = 0$, there are three real eigenvalues. As we increase α , the cubic graph shifts upwards until α reaches some critical value α_c , when the local minimum of the cubic goes above the x -axis, after which there is one real eigenvalue and two complex eigenvalues. In the squirrel (FK) problem, we disallowed complex eigenvalues. Should we also disallow them here?

We haven't given names to types of stability in three dimensions; suffice to say that if $\alpha \leq \alpha_c$, we expect $(u(z), v(z))$ to vary monotonically with z , going from $(\beta, 1 - \beta)$ to $(1, 0)$ as z increases. And if $\alpha > \alpha_c$, we expect the solution to oscillate as z increases away from the equilibrium at $(\beta, 1 - \beta)$. The solutions look like fig. 1.

As for the wavespeed, this equilibrium puts no constraints on it. Coupled with the rule that waves travel at their slowest possible value, c^2 turns out to be equal to the value you specified in question (3c).

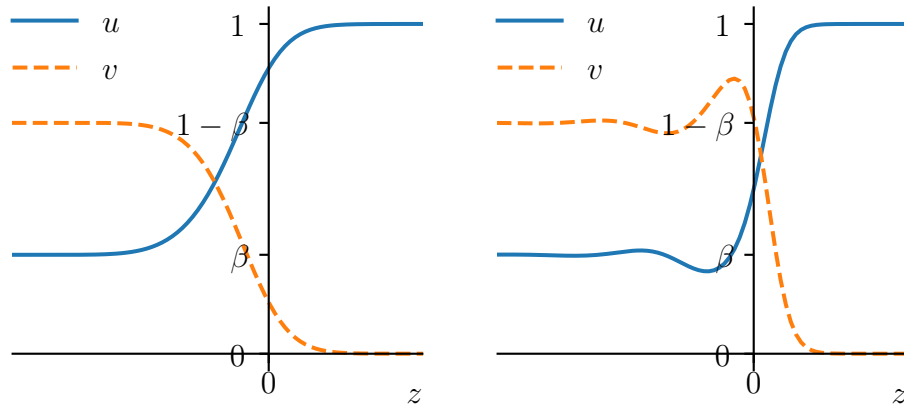


Figure 1: Left: Monotonic approach of (u, v) to $(\beta, 1 - \beta)$ when $\alpha \leq \alpha_c$.
Right: Oscillatory approach to the steady state $(\beta, 1 - \beta)$, when $\alpha > \alpha_c$.